

Communication & Notification Platform Services

Functional Design & Backend Screens

TICVAI | Professional Reference Document

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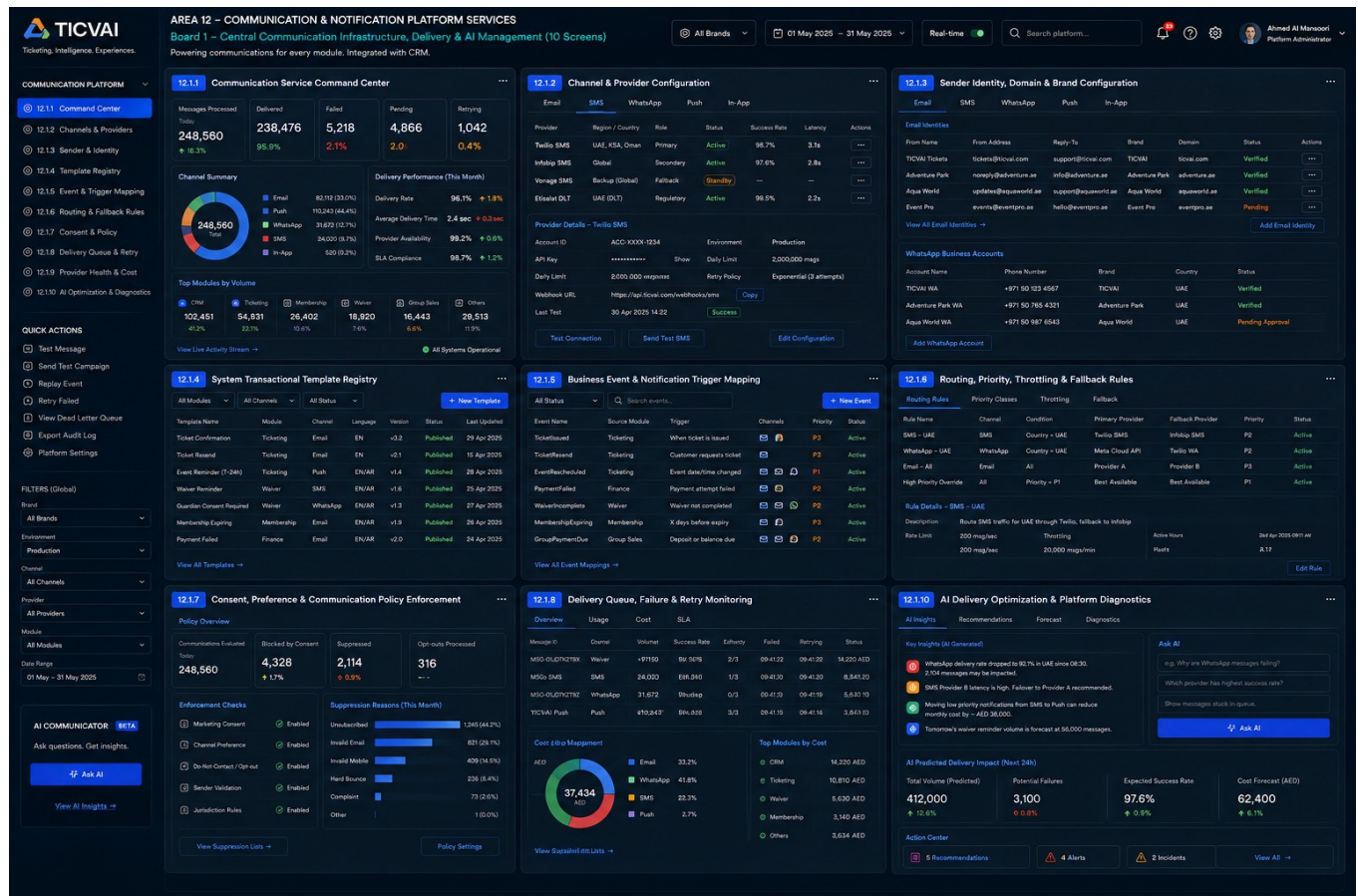
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Area 12 — Communication & Notification Platform Services

Board 1 — Central Communication Infrastructure, Delivery & AI Management



10 Backend Screens

Board Objective

Area 12 should operate as TICVAI's **central communication infrastructure layer**, sitting underneath CRM and all other modules.

It is **not a marketing campaign portal**.

Marketing teams continue to work entirely inside **Marketing & CRM**, while operational modules such as Ticketing, Membership, Waiver, Group Sales, Finance and Access Control use this centralized service whenever they need to communicate.

Core Architecture

TICVAI Business Modules

CRM | Ticketing | Membership | Wallet | Waiver | Group Sales | Finance | Resource Management | Access Control | F&B/Retail



Business Event / Communication Request

↓

TICVAI Communication & Notification Service

↓

Email | SMS | WhatsApp | Push | In-App

↓

Customer / Partner / Employee**The Board 1 flow is:****Monitor → Configure Channels → Configure Identity → Register Templates → Map Events → Route → Govern Consent → Operate Delivery → Monitor Providers → AI Optimize**

Screen 12.1.1 — Communication Service Command Center

Purpose

Provide administrators and technical/operations teams with a centralized view of the health and activity of TICVAI's communication infrastructure.

This is **not** a marketing dashboard.

Platform KPIs

Display:

- Messages Processed Today
- Email Sent
- SMS Sent
- WhatsApp Sent
- Push Notifications
- In-App Notifications
- Delivered
- Failed
- Pending
- Retrying
- Average Delivery Time
- Provider Availability

Module Activity

Show communication volume originating from:

- CRM
- Ticketing
- Membership
- Waiver
- Group Sales

- Customer Service
- Finance
- Wallet
- Resource Management
- Access Control
- Other TICVAI Services

Channel Health

Example:

Channel	Provider	Volume	Success	Latency	Health
Email	Provider A	82K	99.4%	1.8s	Healthy
SMS	Provider B	24K	98.7%	3.1s	Healthy
WhatsApp	Provider C	31K	97.9%	2.4s	Warning
Push	TICVAI Push	110K	99.8%	0.6s	Healthy

Live Operational Alerts

Examples:

WhatsApp delivery failure increased to 8.4% during the last 15 minutes.

1,820 waiver reminders are queued for delivery.

SMS Provider B is approaching its configured rate limit.

Drill-Down

Allow:

Module → Event → Channel → Provider → Message/Transaction

AI Capability

AI should summarize platform health:

WhatsApp performance is degraded in the UAE. Approximately 2,100 pending messages may be affected. Email fallback is available for 78% of recipients.

Acceptance Condition

Platform administrators can understand communication volume, delivery health, provider performance and operational exceptions from one centralized workspace.

Screen 12.1.2 — Channel & Provider Configuration

Purpose

Configure the external/internal services used by TICVAI to deliver communications.

Supported Channels

Configure:

- Email
- SMS
- WhatsApp
- Mobile Push
- In-App Notification
- Future supported channels

Provider Configuration

For each provider:

- Provider Name
- Channel
- Account
- Region
- Environment
- Credentials/Secret Reference
- API Configuration
- Webhook Configuration
- Rate Limits
- Timeout
- Retry Policy
- Priority
- Status

Sensitive credentials should use TICVAI's secure secrets-management architecture rather than be exposed directly in the UI.

Multiple Providers

Support:

Primary Provider

Secondary Provider

Emergency/Fallback Provider

Example:

SMS:

Provider A — Primary

Provider B — Fallback

Regional Configuration

Allow different providers by:

- Country
- Region
- Brand
- Legal Entity
- Environment

Provider Testing

Actions:

- Test Connection
- Send Test Message
- Validate Credentials
- Test Webhook
- Verify Delivery Receipt

Provider Status

Active / Standby / Degraded / Suspended / Disabled

Acceptance Condition

Authorized administrators can configure and test multiple communication providers without requiring individual TICVAI modules to maintain direct provider integrations.

Screen 12.1.3 — Sender Identity, Domain & Brand Configuration

Purpose

Manage the identities from which TICVAI communications are sent.

Email Identity

Configure:

- Sending Domain
- From Name
- From Address
- Reply-To
- Brand
- Legal Entity
- Region

SMS Identity

Configure:

- Sender ID
- Country
- Brand
- Approved Use
- Provider
- Status

WhatsApp Identity

Configure/reference:

- Business Account
- Phone Number
- Brand
- Country
- Provider
- Verification Status
- Approved Templates

Push Identity

Configure:

- Application
- Brand
- Platform
- Environment

Multi-Brand Support

Example:

TICVAI Tenant A

may operate:

- Venue Brand A
- Attraction Brand B
- Event Brand C

Each can use the correct sender identity.

Validation

TICVAI should prevent a module from using an unauthorized sender.

Example:

Sender Brand-A cannot be used for Venue B.

Status

Pending Verification / Verified / Active / Suspended / Expired

AI Capability

AI can detect configuration anomalies such as unusual bounce rates associated with a specific sender identity.

Acceptance Condition

Every outbound communication uses a verified and appropriately governed sender identity for the applicable brand, channel and region.

Screen 12.1.4 — System Transactional Template Registry

Purpose

Maintain centralized **system/transactional communication templates** used by TICVAI operational modules.

This screen must **not replace CRM's marketing template builder**.

Template Examples

Ticketing:

- Ticket Confirmation
- Ticket Resend
- Event Reminder
- Event Rescheduled
- Event Cancelled

Payments:

- Payment Successful
- Payment Failed
- Refund Processed

Membership:

- Membership Activated
- Membership Expiring

Waiver:

- Waiver Required
- Guardian Consent Required
- Waiver Reminder

Group Sales:

- Group Booking Confirmed
- Deposit Due

Template Record

Configure:

- Template ID
- Template Name

- Business Event
- Source Module
- Channel
- Brand
- Language
- Version
- Status

Template Content

Support:

- Subject
- Header
- Body
- Footer
- CTA
- Attachments where applicable

Dynamic Variables

Examples:

{{CustomerName}}

{{OrderNumber}}

{{EventName}}

{{EventDate}}

{{TicketLink}}

{{AmountDue}}

Template Ownership

Clearly identify:

Transactional Template — Platform

versus

Marketing Template — CRM

Versioning

Published templates must be version-controlled.

Acceptance Condition

Operational modules can use governed reusable communication templates without embedding message content directly in application code.

Screen 12.1.5 — Business Event & Notification Trigger Mapping

Purpose

Map TICVAI business events to the operational communications they should generate.

This is the core of the **event-driven communication architecture**.

Event Registry

Examples:

OrderConfirmed

TicketIssued

TicketRescheduled

EventCancelled

PaymentFailed

RefundCompleted

MembershipExpiring

WaiverIncomplete

GroupPaymentDue

ResourceAssignmentChanged

Event Source

Each event should identify:

- Event ID
- Source Module
- Event Type
- Trigger
- Payload
- Priority
- Status

Event-to-Communication Mapping

Example:

Event

TicketIssued

↓

Customer Conditions

Valid email + Email permitted

↓

Template

Ticket Confirmation

↓

Channel

Email

**Send**

Immediately

Multi-Channel Trigger**Example:**

EventCancelled

→ Email

→ Push

→ SMS if event occurs within 24 hours

Scheduled Trigger**Example:**

MembershipExpiring

T-30 days

→ Email

T-7 days

→ Push

T-1 day

→ Email

Trigger Conditions**Allow conditions based on:**

- Product
- Event
- Venue
- Brand
- Customer
- Channel
- Time
- Transaction Status
- Membership
- Booking Type

Important CRM Boundary

CRM marketing journeys should send requests through this service but remain configured and owned within CRM.

Acceptance Condition

TICVAI modules can publish business events while the Communication Service determines the appropriate operational notification according to centrally configured rules.

Screen 12.1.6 — Routing, Priority, Throttling & Fallback Rules

Purpose

Determine **how TICVAI delivers a message after a communication requirement has been created.**

Routing Dimensions

Configure based on:

- Channel
- Country
- Brand
- Provider
- Message Type
- Priority
- Recipient Type
- Cost
- Provider Health

Priority Classes

I recommend:

P1 — Critical

Event cancellation, security or urgent operational communication.

P2 — High

Payment issue, event-day notification.

P3 — Normal

Ticket confirmation, waiver reminder.

P4 — Bulk

Non-urgent high-volume communication.

Provider Routing

Example:

SMS UAE

Primary → Provider A

If unavailable → Provider B

Channel Fallback

Where business rules and consent permit:

WhatsApp Failed

↓

SMS

↓

Email**Throttling****Configure:**

- Messages per second
- Messages per minute
- Provider limit
- Brand limit
- Event limit

Queue Priority

Critical operational messages should not be delayed behind large non-urgent message volumes.

Cost-Aware Routing

Where appropriate, route through the most efficient provider that meets required service and compliance rules.

AI Capability**AI can recommend routing adjustments based on:**

- Success rate
- Latency
- Cost
- Region
- Provider health

Acceptance Condition

Communication requests are dynamically routed according to priority, availability, consent, provider health and configured fallback rules.

Screen 12.1.7 — Consent, Preference & Communication Policy Enforcement

Purpose

Create a central enforcement layer ensuring communications respect the appropriate communication rules.

Critical Architectural Principle

CRM can own customer communication preferences and marketing consent.

But the Communication Service should **enforce applicable policy at delivery time.**

This creates an important separation:

CRM = preference/consent source

Communication Service = enforcement gate

Preference Inputs

Consume:

- Marketing Consent
- Email Preference
- SMS Preference
- WhatsApp Preference
- Push Preference
- Language
- Opt-Out
- Suppression
- Contact Restrictions

Communication Classification

Every communication should be classified, for example:

- Transactional
- Operational
- Service
- Marketing

Policy Evaluation

Before sending:

Communication Type

-

Customer Preference

-

Consent

-

Channel

-

Jurisdiction/Policy

→ Send / Block / Use Alternative Channel

Suppression Lists

Support/reference:

- Unsubscribed
- Invalid Email
- Invalid Mobile

- Hard Bounce
- Complaint
- Administrative Suppression

Audit**Record why a communication was:**

- Allowed
- Blocked
- Rerouted
- Suppressed

Acceptance Condition

No communication is delivered without passing the applicable centralized consent, preference and communication-policy evaluation.

Screen 12.1.8 — Delivery Queue, Failure & Retry Management

Purpose

Provide technical/operations teams with visibility into communications currently being processed or failing.

Queue Dashboard**Display:**

- Pending
- Processing
- Sent
- Delivered
- Failed
- Retrying
- Dead-Lettered
- Cancelled

Message Record**Display:**

- Communication ID
- Source Module
- Business Event
- Recipient
- Channel
- Template
- Provider

- Priority
- Created
- Sent
- Status
- Attempts

Sensitive recipient information should be masked according to role.

Failure Categories

Examples:

- Provider Unavailable
- Invalid Address
- Invalid Mobile
- Rate Limited
- Authentication Error
- Template Rejected
- Timeout
- Consent Block
- Unknown Error

Retry Policy

Configure:

Attempt 1

↓

Wait 1 minute

↓

Attempt 2

↓

Wait 5 minutes

↓

Attempt 3

↓

Fallback Provider / Dead Letter

Operations Actions

Authorized administrators can:

- Retry
- Cancel
- Reroute
- Change Provider

- Inspect Failure
- Replay Event where safe

Dead-Letter Queue

Failed messages requiring intervention should move into a controlled exception queue.

AI Capability

AI can group failures by probable root cause.

Example:

4,821 SMS failures originate from Provider B authentication errors beginning at 08:14.

Acceptance Condition

Failed communications can be identified, diagnosed, retried and resolved without losing the original business-event context.

Screen 12.1.9 — Provider Health, Usage & Cost Monitoring

Purpose

Monitor the operational and commercial performance of communication providers.

Provider KPIs

Display:

- Volume
- Success Rate
- Failure Rate
- Delivery Time
- API Latency
- Availability
- Retries
- Fallback Usage
- Cost
- Cost per Message

Provider Comparison

Example:

Provider	Success	Avg. Delivery	Cost/1K	Health
SMS A	99.1%	2.2 sec	AED X	Healthy
SMS B	97.4%	4.8 sec	AED Y	Warning

Usage Analysis

Analyze by:

- Channel
- Provider
- Brand
- Venue
- Module
- Country
- Message Type

Cost Allocation

Communication costs can be attributed to:

- Tenant
- Brand
- Venue
- Module
- Campaign where CRM provides the reference
- Event
- Business Unit

Threshold Alerts

Examples:

SMS monthly spend has reached 85% of configured budget.

WhatsApp Provider A success rate dropped below 97%.

SLA Monitoring

Track contractual/provider targets where configured.

AI Capability

AI can recommend provider changes based on cost/performance trade-offs.

Acceptance Condition

Management can understand provider reliability, usage and communication cost across the TICVAI platform.

Screen 12.1.10 — AI Delivery Optimization & Communication Platform Diagnostics

Purpose

Provide an AI intelligence layer focused specifically on **communication infrastructure performance**, not CRM marketing strategy.

AI Inputs

Analyze:

- Business Events
- Channels
- Providers
- Queues
- Failures
- Retries
- Latency
- Delivery
- Cost
- Recipient preferences
- Regional performance
- Historical patterns

Natural-Language Queries**Administrators can ask:**

“Why are WhatsApp messages failing today?”

“Which provider is performing best in the UAE?”

“Which modules generated the most SMS this month?”

“Why did communication cost increase?”

“Which messages are currently stuck?”

AI Incident Detection**Example:****Potential Provider Incident**

SMS Provider B failure rate increased from 1.2% to 14.7% during the last 12 minutes.

3,842 messages affected.

Provider A has sufficient capacity for failover.

AI Routing Recommendation

Move UAE transactional SMS to Provider A until Provider B success rate exceeds 98% for 15 consecutive minutes.

Cost Optimization**Example:**

Moving eligible low-priority notifications from SMS to Push could reduce estimated monthly communication cost by AED 38,000.

Only where customer/channel rules permit.

Volume Forecasting**Forecast communication load for:**

- Major Events

- Ticket Releases
- Membership Renewals
- Group Arrivals
- Waiver Deadlines

AI Root-Cause Analysis

AI should connect:

Problem → Provider → Region → Message Type → Source Module

rather than merely reporting that messages failed.

Automation Governance

Approved low-risk actions could eventually include:

- Provider Failover
- Queue Scaling
- Retry Adjustment
- Operational Alerting

But changes affecting customer consent, marketing communication or message meaning must remain governed by their owning modules/policies.

Acceptance Condition

TICVAI provides explainable AI diagnostics and recommendations that improve communication reliability, scalability and cost without interfering with CRM's ownership of customer marketing strategy.

Board 1 — Final Screen Register

Screen	Backend Screen	Primary Responsibility
12.1.1	Communication Service Command Center	Platform communication health
12.1.2	Channel & Provider Configuration	Provider/channel setup
12.1.3	Sender Identity, Domain & Brand Configuration	Sender governance
12.1.4	System Transactional Template Registry	Operational templates
12.1.5	Business Event & Notification Trigger Mapping	Event-driven notifications
12.1.6	Routing, Priority, Throttling & Fallback Rules	Delivery orchestration
12.1.7	Consent, Preference & Communication Policy Enforcement	Communication governance

Screen	Backend Screen	Primary Responsibility
12.1.8	Delivery Queue, Failure & Retry Management	Delivery operations
12.1.9	Provider Health, Usage & Cost Monitoring	Provider performance/cost
12.1.10	AI Delivery Optimization & Platform Diagnostics	AI infrastructure intelligence

Area 12 — Final Architecture

I would keep **Area 12 at only 1 board / 10 screens**.

The ownership should be very clear:

Marketing & CRM owns

Audience → Segmentation → Campaign → Journey → Marketing Content → Marketing Schedule → Marketing Analytics

Area 12 owns

Provider → Sender → Transactional Template → Business Event → Routing → Consent Enforcement → Queue → Delivery → Failure → Infrastructure Analytics

Other TICVAI modules own the business reason

For example:

Waiver

"Guardian consent is incomplete."

↓

Area 12

Determines the approved template, channel, provider and delivery process.

↓

Customer

And for CRM:

CRM

"Send this campaign to this approved audience."

↓

Area 12

Validates delivery eligibility and handles channel/provider delivery.

↓

Customer

This means a marketing user **never needs to enter Area 12**, while the technical/platform team has one powerful central communication infrastructure underneath the entire TICVAI ecosystem.

That is the architecture I would lock into the SOW.